

CMP4272- DATA STRUCTURES AND ALGORITHMS

LAB -7: Algorithm Plans

Objectives:

- Specify computational problems by their required inputs and outputs.
- Develop your repertoire of "algorithmic plans" - building blocks which each solve "part" of a problem.
- Select and compose "algorithmic plans" to write algorithms for simple computational problems.

1. Background:

An algorithm is an explicit, precise, unambiguous, finite sequence of elementary instructions, usually intended to accomplish a specific purpose.

Characteristics of an algorithm

1. Finite: An algorithm must terminate after a finite number of steps.
2. Definite: Each step in the algorithm must be precisely and unambiguously defined.
3. Effective: Algorithms must be executable and practical, meaning they can be carried out using available resources within a reasonable amount of time.
4. Input/Output: An algorithm takes zero or more inputs and produces one or more outputs.
5. Correctness: An algorithm should produce the correct output for all valid inputs.
6. Deterministic: Given the same input, an algorithm should always produce the same output.

2. Complete the following exercises.

Exercise-1:

Consider the problem statement and a proposed algorithm for the given problem.

Problem statement

The problem is to maintain a comfortable indoor temperature in a building by controlling the operation of the central heating system.

Proposed algorithm (Pseudo-code)

```
Procedure ControlHeatingSystem():  
    Read current indoor temperature from sensors  
    If indoor temperature is below lower threshold:  
        Activate heating system  
    Else if indoor temperature is above upper threshold:  
        Deactivate heating system
```

Identify ambiguity in the proposed algorithm and discuss how to address it.

Answer:

Ambiguity:

The Algorithm does not define the lower and upper temperature value, and also does not define what happens when the temperature is between both thresholds.

Address it:

Clearly define the upper and lower temperature limits.

Exercise-2: Name the plans used in the following code, and specify the problem this function solves.

i)

```
def exercise2(numbers):  
    total = 0  
    for value in numbers:  
        cube = value ** 3  
        total += cube  
    return total
```

Answer:

Plans: do something for each value

:Map

:running total

Problem solution: sum of the cube of numbers.

ii)

```
def exercise3(numbers, cutoff):  
    for i, value in enumerate(numbers):  
        if value >= cutoff:  
            return i  
    return -1
```

Answer:

Plan :do something for each value

:Filter

:Linear Search

Problem solution: find the index of the first number greater than or equal to the cutoff.

Exercise-3: For each of the problem statements below:

- Highlight the key words to determine the inputs and outputs.
- Identify the subproblems.
- Select plans to solve the subproblems.
- Write a function to solve the problem, by combining the plans in code.
- Test your solution with example inputs.

i) Find the sum of a list of numbers.

Answer:

Key words: sum, list of numbers

Sub problem 1: check each number.

Sub problem 2: add it to the total.

Plans: running total, do something for each value

ii) Count how many even values are in a list of integers.

Answer:

Key words: even numbers, list of integers

Sub problem 01:check each values

Sub problem 02:count even values

Plans: counter,filter, do something for each value

iii) Find the maximum value from a list of numbers.

Answer:

Key words: list of number, maximum value

*Sub problem: compare each number with other number.
Plans: counter, do something for each value, best so far*

- iv) Given a list of strings, find the sum of their lengths.

Answer:

Key words: list of strings, sum, lengths

Sub problem 01: find the length of each string

Sub problem 02: add lengths

Plans: Running total, Map

- v) Given a list of strings, find the index of the shortest one.

Answer:

Key words: list of strings, index, shortest one

Sub problem 01: find out the index

Sub problem 02: compare each number

Sub problem 03: find out the shortest one

Plans: linear search, find so far, do something for each value